

(1)

(a) The synchronous speed is 1800 rpm

$$(b) s = \frac{1800 - 1720}{1800} = 0.044$$

$$(c) \text{ Rotor current freq.} = (1800 - 1720) \times \frac{2}{60} = 2.667 \text{ Hz}$$

(d) @ full load, the speed of the rotating field produced by the rotor current:

$$\text{w.r.t. the rotor is } 1800 - 1720 = 80 \text{ RPM}$$

$$\text{w.r.t. the stator structure is } 1800 \text{ RPM}$$

(e) @ standstill, the speed of the rotating field produced by the rotor current:

$$\text{w.r.t. the rotor is } 1800 \text{ RPM}$$

$$\text{w.r.t. the stator structure is } 1800 \text{ RPM}$$

(2) The induction motor may run at synchronous speed if under ideal conditions that is frictionless and no load. It is impossible in practice since there is always friction.

(3) Rotational losses:

→ loss due to friction and windage

(4) the tests are to find the parameters of the motor.

(5) At no-load condition, the slip is near zero.